

Our CIE program is also committed to achieving criteria for Accrediting Engineering Programs defined by ABET, which include, as part of Criterion 5,

"a culminating major engineering design experience that 1) incorporates appropriate engineering standards and multiple constraints, and 2) is based on the knowledge and skills acquired in earlier course work."

The engineering design experience as defined by ABET is:

"a process of devising a system, component, or process to meet desired needs and specifications within constraints. It is an iterative, creative, decision-making process in which the basic sciences, mathematics, and engineering sciences are applied to convert resources into solutions. Engineering design involves identifying opportunities, developing requirements, performing analysis and synthesis, generating multiple solutions, evaluating solutions against requirements, considering risks, and making trade-offs, for the purpose of obtaining a high-quality solution under the given circumstances. For illustrative purposes only, examples of possible constraints include accessibility, aesthetics, codes, constructability, cost, ergonomics, extensibility, functionality, interoperability, legal considerations, maintainability, manufacturability, marketability, policy, regulations, schedule, standards, sustainability, or usability."

Therefore, your final report is only eligible if it has the following elements:

Problem

1. A clear problem definition, motivation and scope that respond to identified opportunity.

Requirements

2. A clear definition of the requirements.
3. A clear definition of the constraints such accessibility, aesthetics, codes, constructability, cost, ergonomics, extensibility, functionality, interoperability, legal considerations, maintainability, manufacturability, marketability, policy, regulations, schedule, standards, sustainability, or usability.
4. A clear definition of the engineering standards that are used in the design and implementation of the project (please note that your report shall include clear evidence of the incorporation of these standards in your design and implementation).
5. A clear definition of the risks related to the project
6. A clear definition of the laws and regulations affecting your design.

Design

7. A discussion of multiple solutions you generated to respond to the identified problem(s).
8. A discussion of the decision analysis technique you used to select a proper design.
9. A discussion of how you evaluated your solution against the requirements, constraints, and which trade-offs you had to make to respond to the risks.

Implementation

10. A description of your initial implementation.
11. A description of the iterative process you used to improve your implementation.
12. A description of the final implementation.

Impact

13. A discussion of how you considered the impact of your project on the public health, safety and welfare.
14. A discussion of how the global factors related to your project.
15. A discussion of the economic factors related to your project.
16. A discussion of the cultural and social factors related to your project.
17. A discussion of the environmental factors related to your project.